

Literature Review

1. Overview

E-commerce platforms, which allow users to conveniently browse, choose, and buy things online, have become an essential component of contemporary digital environments. To guarantee a flawless user experience, these solutions prioritize usability, performance, and accessibility. Successful e-commerce platforms combine technology and user-centered design to enhance customer pleasure and corporate performance, according to E-commerce 2023: corporate, Technology, Society.

2. Analysis of Similar Systems

2.1 Amazon

One of the most sophisticated e-commerce sites is Amazon, which has features like customized suggestions, sophisticated search filters, and a quick checkout procedure. By saving time and effort, these features greatly improve the user experience. Personalization and effective navigation are crucial elements in enhancing e-commerce platform usability, according to Nielsen Norman Group.

2.2 At noon

With an emphasis on accessibility and simplicity for users in the Middle East, Noon offers a clear and easy-to-use interface. Its design places a strong emphasis on obvious offers and unambiguous product classification. Users favor systems that offer simple access to discounts and bargains as well as intuitive navigation, according to Statista research._

2.3 Jumia

Jumia offers a streamlined shopping experience with support for regional payment options, focusing on emerging markets. Despite having a simpler interface, it is nevertheless useful for novice users. Design simplicity is crucial for boosting adoption in developing countries, according to Statista.

3. Feature Comparison

There are variations in performance, usability, and personalization when comparing the main e-commerce systems. While Noon and Jumia place a higher priority on accessibility and simplicity, platforms like Amazon place a strong emphasis on personalization. E-commerce 2023: Business, Technology, Society states that the success of e-commerce systems depends on striking a balance between usability and functionality.

4. User Experience (UX) Insights

For e-commerce systems to succeed, user experience is essential. Easy navigation, quick loading times, responsive design, and a streamlined checkout procedure are important components. Reducing the processes needed to finish a purchase greatly increases customer satisfaction and conversion rates, according to Nielsen Norman Group. Furthermore, Google Developers emphasizes the importance of speed optimization, pointing out that faster websites result in higher engagement.

5. Application to Our Project

This project uses Bootstrap and React to provide a clear and responsive user interface based on the analysis. Product data will be efficiently displayed by the system after being retrieved from an external REST API. To guarantee a seamless and long-lasting user experience, local storage will be used to handle the shopping cart. These design choices are in line with contemporary e-commerce development best practices.

6. Feedback & Evaluation

The project shows a thorough grasp of e-commerce principles and the application of contemporary front-end technology. Scalability and maintainability are enhanced by the usage of React and component-based architecture. However, the system's completeness is limited by the lack of security measures and backend integration. E-commerce 2023: Business, Technology, Society states that user authentication and safe transactions are essential components of a complete e-commerce system.

7. Suggested Improvements

To enhance the system, several improvements can be implemented:

- Enhance the functionality of search and filtering
- Include ratings and reviews for products.
- Include safe payment methods
- Product Comparison Feature
- Smart Cart System (Auto Suggestions)
- Dark Mode Toggle

8. Final Grading Criteria

The project can be evaluated based on the following criteria:

- Quality of Documentation (20%)
- Code quality and implementation (30%)
- Functionality (API Integration, Cart) (20%)
- Debugging and Testing (10%)
- User Experience and Interface (10%)
- presentation (10%)

The evaluation methods used in software engineering projects are reflected in these criteria (Laudon & Traver, 2023).
